Inner-ring pixel history audit

Completed October 14, 2026. Descriptive spatial analysis, exclusive 0-25 miles, 2013-2025. No econometric estimates. Workflow 38515721688 completed successfully.

Coverage and validation

All 57 nonempty catchments have 13 years and 12 adjacent-year transition tables. The 58th candidate has no nonempty exclusive inner catchment. Annual class totals equal coverage counts. Endpoint transitions and persistence sum to complete-series coverage. Peak-cohort sizes are constant across observed years. All 496 georeferenced tiles passed CRS, 30 m resolution, band, persistence, soybean-entry and endpoint-forest count checks.

Complete-series support: 293,323,652 pixels (65,233,668 acres). Fixed nonwater support: 282,174,631 pixels (62,754,183 acres). Annual valid coverage increases in 2025; complete-series histories exclude newly observed pixels. Pairwise annual changes and complete-period histories therefore have different supports. Water is retained as a transition category, not silently treated as new land.

Persistent soybeans versus rotation

164,257,754 pixels were pure soybeans at least once on complete-series support. Only 293,904 were soybeans in every year (0.18% of ever-soy pixels, 65,363 acres). 48.52% of ever-soy pixels were soybeans in six or seven of the 13 years. This is compatible with rotations; an always-soy criterion is not a sensible measure of a permanent soybean-growing region. Rotation cannot be conclusively diagnosed from frequency alone.

Pixels soybean in 2025 but nonsoy in 2013 came principally from corn (63.70% of that endpoint cohort). This is an endpoint classification comparison, not proof of permanent expansion or first-ever soybean cultivation. The retrospective figure also shows the prior-year labels of peak-year soybean cohorts, using each hub's own peak year. 1 hubs peak in 2013 and have no preceding year in this dataset. Pure soy is class 5; double-cropped soybean categories are retained separately.

Forest: where did new labels come from?

In 2019 there were 8,441,523 gross forest-gain pixels and 4,293,725 forest-loss pixels on pairwise common support. Net gain: 4,147,798 pixels (922,449 acres). The largest origins are shrub/scrub (30.64%), grass/pasture (30.36%), woody wetlands (15.63%) and developed open space (14.27%). Soybeans account for only 0.80% of gross forest gains.

Forest labels show net losses in 2021 and 2025, not uninterrupted forest growth. The annual subtype figure shows a pronounced shift toward evergreen/mixed forest in 2019. Earlier USDA metadata checks documented updated NLCD ancillary inputs, making classifier/source changes a plausible explanation for part of these jumps. These tables do not quantify how much is physical land change. Independent imagery or a consistent NLCD time series is necessary before claiming afforestation, deforestation or clearing.

Top 2019 net forest-gain hubs and largest 2025 loss hubs are retained in forest_annual_net_by_hub.csv. The atlas maps every catchment. Display images use 120 m nearest-neighbor mosaics for readability; all validation and tables use the original 30 m maps. Endpoint maps show 2013 versus 2025, not cumulative clearing across all years.

Locations of the largest forest-label changes

The largest 2019 net gains occur around the inventory sites in Cofield, NC (hub 59), Fayetteville, NC (58), Valdosta, GA (57), Gainesville, GA (48), and Guntersville, AL (35). The five together account for 51.90% of pooled net gains. This concentration in southeastern catchments complements the subtype/source-update evidence, but it does not establish the share of changes caused by classification. Facility names and addresses come from the legacy inventory and do not resolve all site-role questions. The geographic figure labels the three largest absolute changes in each year by hub ID.

Other pathways and the external comparison

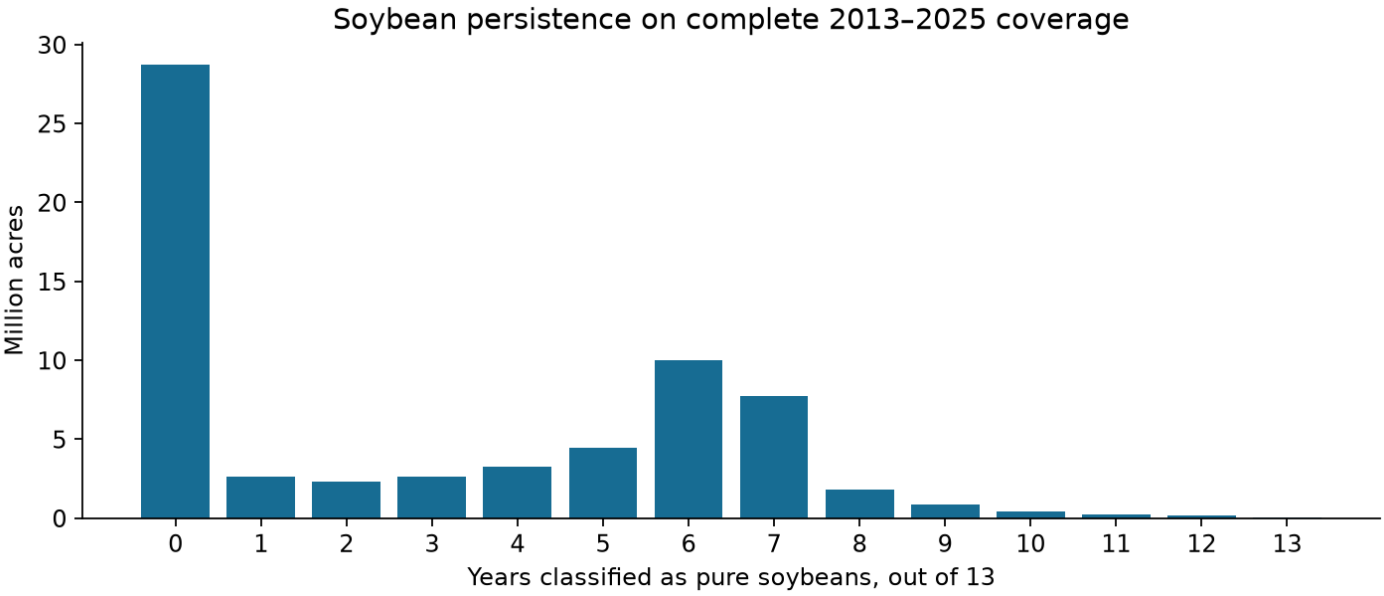
Other LCFS pathways could affect soybean through substitution among oils, competing acreage, and wider commodity prices. The sign can be positive or negative. It is a future sensitivity check, not a demonstrated explanation for our smaller estimate. Avoid counting the same refinery capacity multiple times across feedstock approvals. Capacity is potential demand; actual soybean-based fuel volume is a different treatment. Constant utilization/feedstock proportions alone do not mechanically change a percentage elasticity. Miao, Li and Khanna also differ in period, geography, outcomes and identification. Common national price effects can be absorbed by our year fixed effects.

Sources and reproducibility

- Raw counts: output/pixel_history_inner, generated by code/88_audit_inner_ring_pixel_history.py.
- Workflow: https://github.com/pliedo/lcfs-soybean-land-use/actions/runs/38515721688
- Forest-input evidence: doc/forest-classification-evidence.md; USDA Iowa metadata 2018-2021.
- Reproduce this report and figures with code/89_summarize_inner_pixel_history.py and the per-hub workflow map artifacts.
- Fuel comparison: https://onlinelibrary.wiley.com/doi/10.1002/aepp.70005

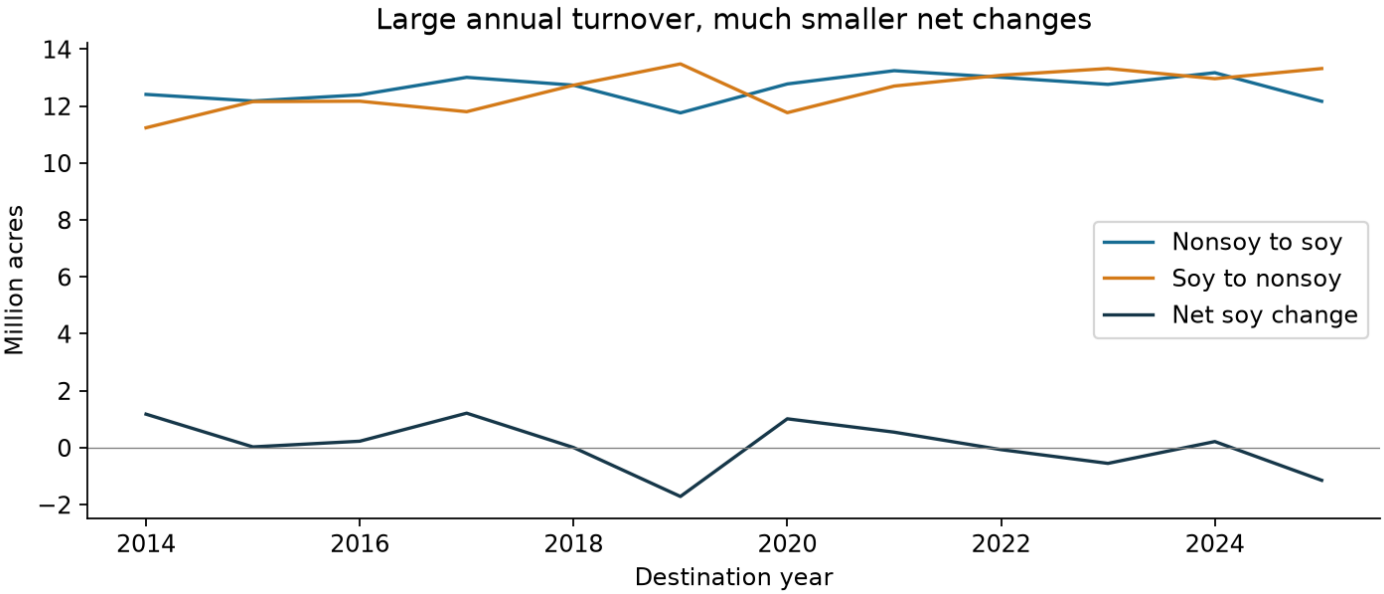
The regression composition figure and this spatial transition audit answer different questions. Neither labels individual transitions as policy-caused.

Soy Persistence



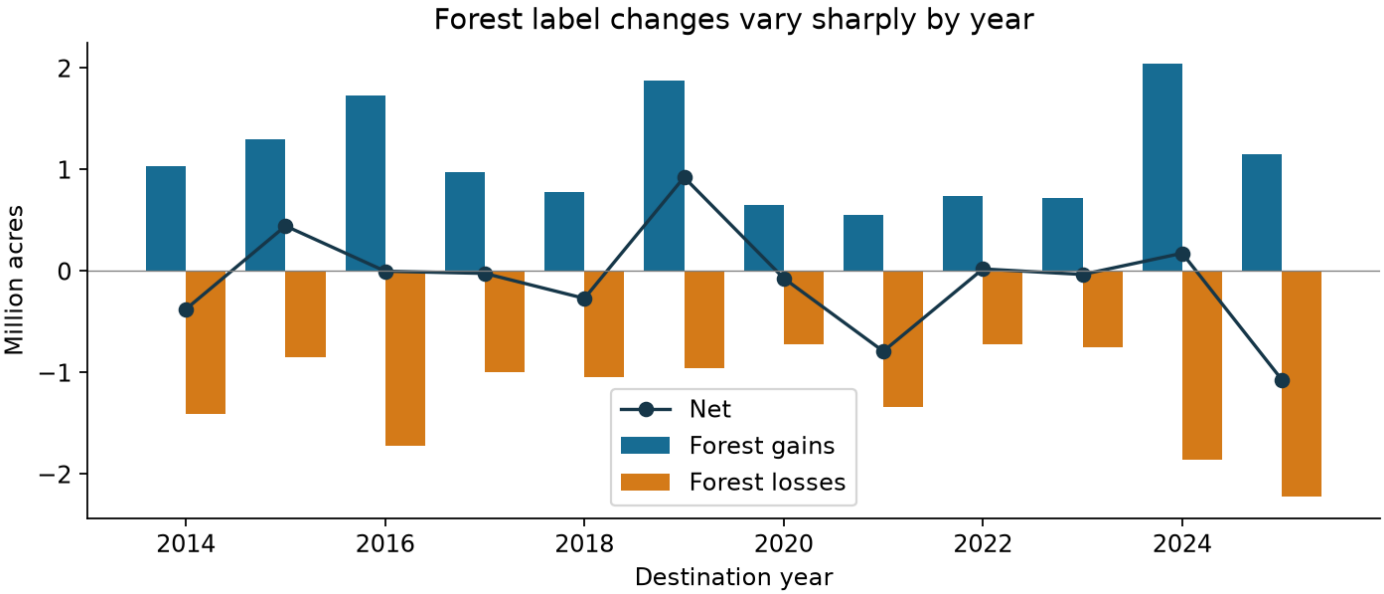
Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

Soy Annual Turnover



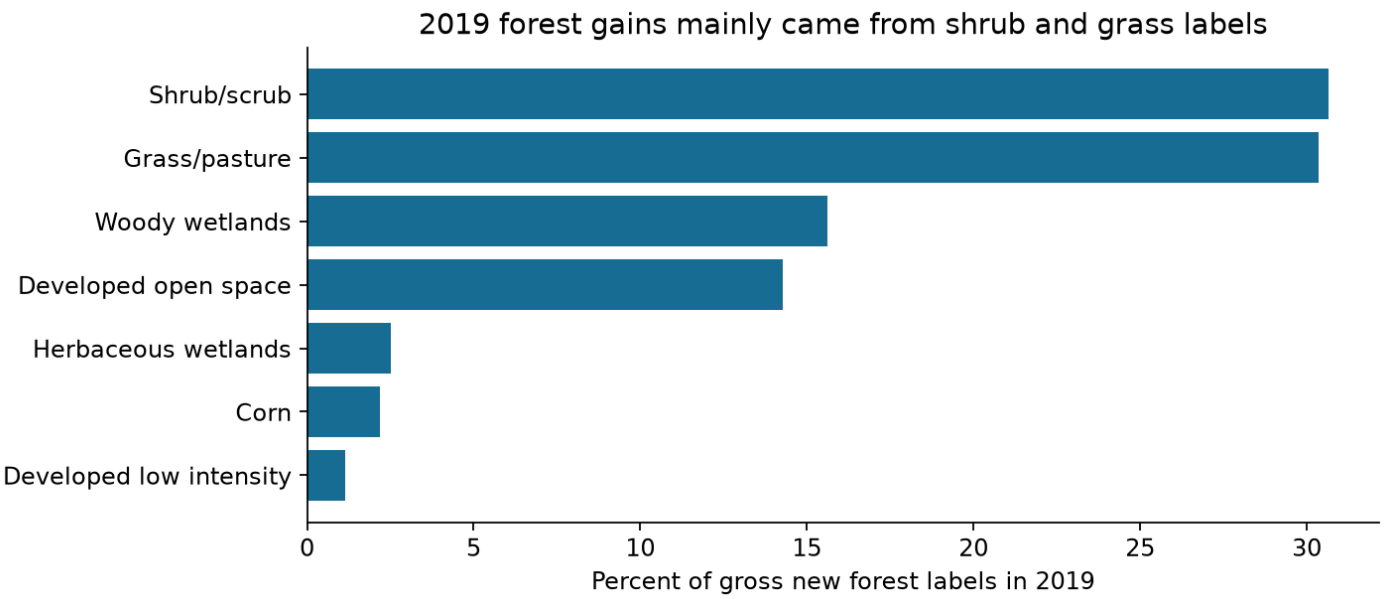
Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

Forest Annual Turnover



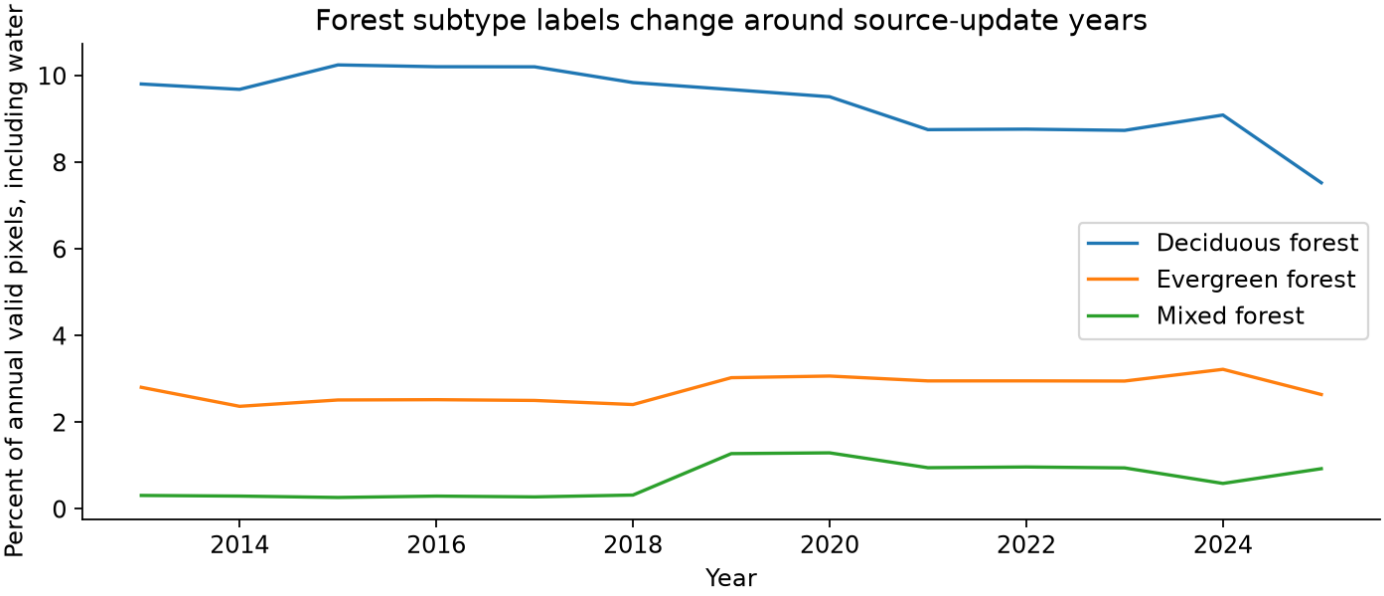
Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

Forest 2019 Sources



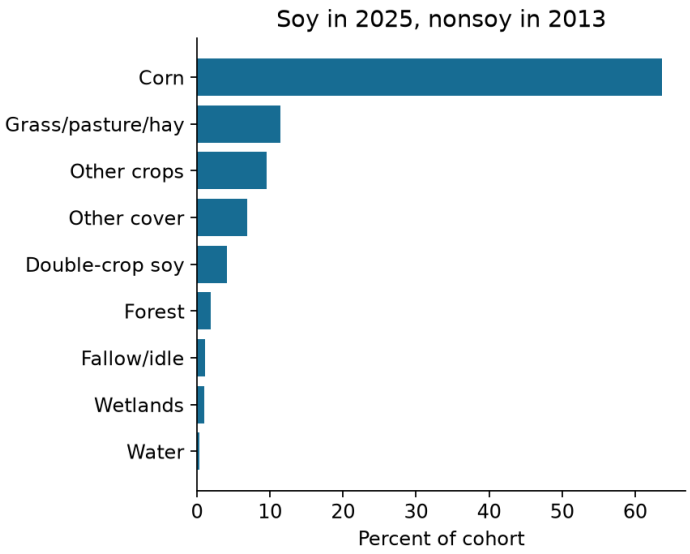
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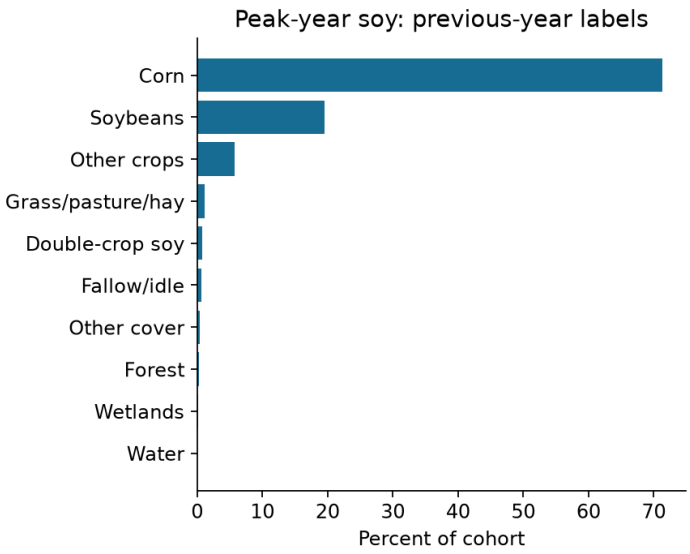
Forest Subtypes



Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

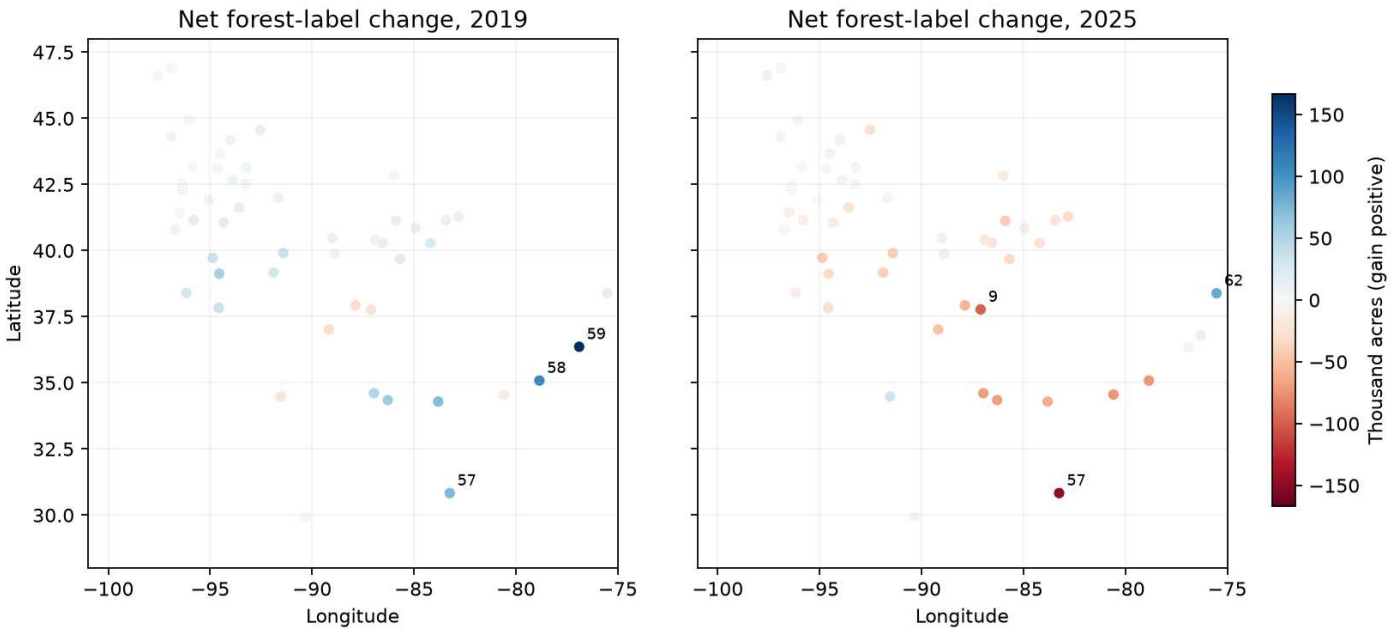
Soy Cohort Origins





Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

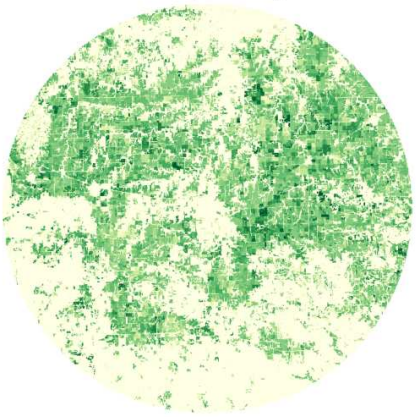
Forest Change Geography



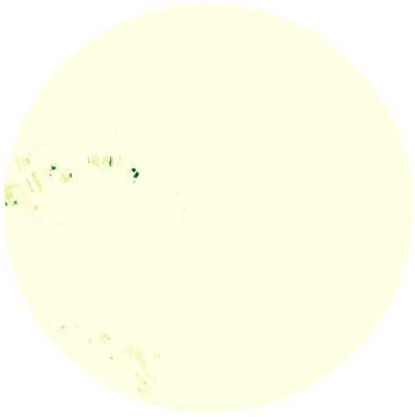
Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.

Example History Maps

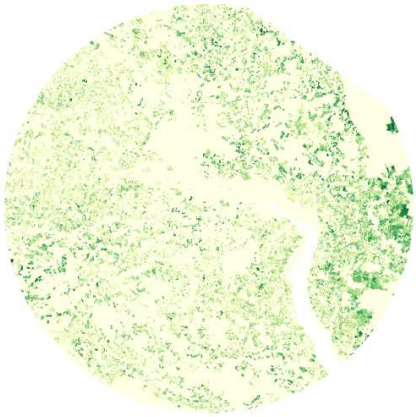
Hub 1: soybean years



Hub 51: soybean years



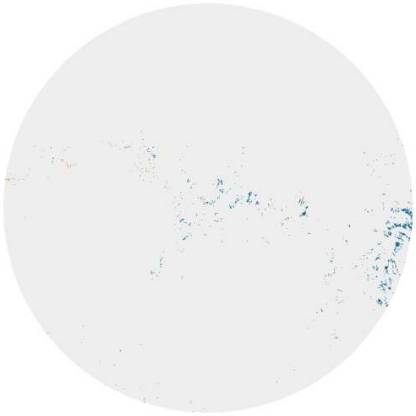
Hub 59: soybean years



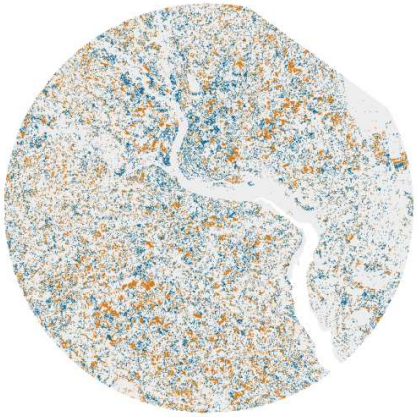
Forest: orange loss / blue gain



Forest: orange loss / blue gain



Forest: orange loss / blue gain



Descriptive CDL labels. No causal attribution. All source counts use 30 m pixels.